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Evidence from the Chilean Voucher System

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ABSTRACT

This work considers a major educational reform in Chile that increased vouchers by 50 percent for students in the lowest 40 percent of the income distribution. This increased the revenues schools received for these students and lowered the relative prices of private voucher schools for eligible parents. I use a national data set to implement a regression discontinuity design (RDD) exploiting that eligibility is a discontinuous function of a socioeconomic ranking. Results reject that eligible students chose schools with higher test scores or socioeconomic status and that these students are doing better than noneligible students in math and language test scores.

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I. Introduction

Advocates of school vouchers typically argue that vouchers help promote consumer choice, personal advancement, and competition. From the advocates' perspective, vouchers lower educational costs even as they increase school quality.¹ Despite this expectation, there is still no consensus in the literature on whether voucher programs improve average outcomes. Moreover, there is concern that vouchers can generate a sorting of students across schools along characteristics such as income and ability, which possibly leads to lower educational outcomes for less-advantaged students. (See Manski 1992; Epple and Romano 1998; and MacLeod and Urquiola 2012, 2015 for theoretical work. See Chakrabarti 2009; Hsieh and Urquiola 2003, 2006; McEwan, Urquiola, and Vegas 2008; and Muralidharan and Sundararaman 2015 for empirical evidence.)

An unanswered question is whether a more careful voucher design could preserve the potential efficiency benefits from competition while mitigating socioeconomic stratification. For instance, several authors have suggested that deviating from a flat voucher to one that conditions the subsidy on student characteristics, such as income, could ameliorate sorting impacts and help low-income students benefit from voucher systems (Epple and Romano 2008; Nechyba 2000, 2003). The underlying idea is that differentiated vouchers would incentivize schools to serve low-income students, which mitigates the schools' temptation to admit only students with relatively high socioeconomic status (SES) or ability. Despite much discussion, there is still little rigorous empirical evidence on the actual impact of differentiated vouchers.

In this work, I look at a major educational reform implemented in Chile in 2008 in which voucher amounts were increased by 50 percent for students in the lowest 40 percent of the income distribution. Chile's school system has had a universal voucher system for more than three decades and also has high levels of socioeconomic stratification, which has been attributed to the voucher system (Hsieh and Urquiola 2003, 2006; McEwan, Urquiola, and Vegas 2008). The introduction of progressive vouchers in 2008 was intended to acknowledge that educating low-income students costs more and to give low-income students greater access to higher-performing voucher schools. To ensure greater student access to higher-performing schools, schools were not just given an incentive to admit low-income students, but, additionally, in exchange for the extra resources, they were forbidden from charging add-on fees to low-income parents (a practice that is common among private voucher schools in Chile). In this work, I estimate the causal effect of students' being eligible for the extra voucher amount and being able to attend a number of higher-performing private voucher schools for free on their school choices and educational outcomes.

Estimating the causal effect that progressive vouchers had on their beneficiaries is challenging because of the endogeneity of progressive voucher access. Despite much research on the topic (MINEDUC 2012; Correa, Parro, and Reyes 2014; Villarroel 2012; Mizala and Torche 2013; Neilson 2013; Navarro-Palau 2017; Feigenberg, Rivkin, and Yan 2019), there is an ongoing discussion of the effects that the Chilean reform has had on its beneficiaries, which partly reflects this identification challenge. I use a new

1. See Hoxby (2003) for a review of how school choice might affect school productivity.

matched administrative data set and exploit that eligibility is a discontinuous function of a socioeconomic ranking to implement a regression discontinuity design (RDD). Although several researchers had noted that it would be natural to assess the program using such an approach, this had not been feasible previously due to a lack of information.

Although the new policy expanded choice sets for eligible students, results show that being eligible for the extra voucher amount did not significantly change students' school choices. Focusing on first-grade students who entered the educational system in 2012—four years after the reform was implemented—I am able to show that being eligible for the extra voucher amount had no impact on the probability of choosing a private school, the test scores of the chosen schools, the SES of the chosen schools, the average class size of the chosen schools, or the distance traveled to school. Results do show that eligible students chose schools that charge higher add-ons to noneligible parents, but this effect is small in magnitude, with eligible students choosing schools that charge approximately \$3 more to noneligible parents.²

Looking at educational outcomes, I show that being eligible for the extra voucher amount had no impact on students' performance on a second grade standardized language test or on a fourth-grade standardized language and math test. Importantly, I am able to reject a positive impact above 0.06 standard deviations on each of these test scores, which reinforces the idea that the policy did not lead parents to choose schools that were better for their children. It also allows me to discard the notion that the extra resources are making a difference for eligible as opposed to noneligible students. The lack of a difference in the test scores of eligible and noneligible students could be a result of schools not differentially providing extra resources to program beneficiaries or resources not being used in a way that improves educational outcomes for eligible students. The fact that eligible students' educational outcomes are not improving any more than noneligible students' outcomes is relevant, considering that the government intended the extra resources to be targeted specifically to eligible students and that the law specifically requires that money from progressive vouchers must be spent on plans to increase the educational results of eligible students.

I perform several exercises intended to provide a better understanding of what could be driving the null effects. I begin by analyzing the extent to which the reform changed choice sets for eligible students. This is relevant considering that, on average, students are not willing to travel major distances to go to school (Hastings and Weinstein 2008; Bayer, Ferreira and McMillan 2007). However, a simple simulation exercise shows that there was substantial room for eligible students to choose better-performing schools. For example, assuming that each student will choose the best-performing school that is free within 0.7 miles (which is in the 50th percentile in terms of distance traveled), I can show that, had this assumption been the criteria, the expectation is that the extra voucher amount and consequent price decrease would allow students to attend schools with test scores roughly 0.4 standard deviations higher and SES 0.35 standard deviations higher. In a second check, I can also show that results remain roughly unchanged when I focus on students for whom the policy should have expanded choice sets the most.

2. All figures are in U.S. dollars.

In a second exercise, I consider whether results are any different for higher- or lower-SES students. Results could differ by SES either because higher-SES parents might have a greater preference for school quality (see Hastings and Weinstein 2008; Bayer, Ferreira, and McMillan 2007) or because schools might choose to serve, among eligible students, those with higher SES. However, I find no evidence of a differential impact on students who have a mother with less than a high school education or a mother with more than a high school education.

In a third exercise, I analyze whether results are specific to 2012. I use data on the socioeconomic ranking for 2008 to test how being eligible for progressive vouchers changed first-grade students' school choices and educational outcomes in 2008 and 2009. Although the data on the socioeconomic ranking for 2008 are much noisier, I find the data still serve as a reasonably good instrument for eligibility in subsequent years. Although more imprecise, results show no effect on students' school choices or educational outcomes.

Finally, I analyze whether results could be driven by the fact that ineligible students who are right above the threshold are likely to become eligible in subsequent years. The socioeconomic ranking used to determine eligibility can change from year to year. Although almost no eligible student loses their status in subsequent years, a number of students who are right above the threshold become eligible in future years. Even though parents have little information on how eligibility is actually determined, I check whether results could be driven by parents of ineligible students close to the threshold foreseeing that their children might become eligible in the future and making better choices. To do so, I implement a matching strategy and compare individuals who are below the threshold with a broader control group that is unlikely to become eligible in the future. To avoid any endogeneity bias driven by unobservables, I closely follow Angrist and Rokkanen (2015) and focus on a narrow group of students for whom I can break the link between the running variable and expected outcomes by conditioning on a relevant set of controls. Results remain the same when using this altered control group.

This work contributes to several aspects of the literature. It contributes to the literature on school vouchers and socioeconomic stratification. Existing studies typically find that voucher systems can lead to increased stratification by SES. See Hsieh and Urquiola (2003, 2006) and McEwan, Urquiola, and Vegas (2008) for evidence on Chile; Sandstrom and Bergstrom (2005), Bohlmark and Lindahl (2007), and Bohlmark, Holmlund, and Lindahl (2015) for evidence on Sweden; and Muralidharan and Sundararaman (2015) for evidence on India. This study shows that progressive vouchers may be ineffective in terms of mitigating these effects.

This work also contributes to the literature on school vouchers and educational outcomes. There is a broad strand of literature analyzing the effects of implementing small- and large-scale voucher systems on educational outcomes, which has typically found mixed results.³ However, little is known about the effects that progressive vouchers have

3. In small-scale voucher systems, we expect vouchers to affect educational outcomes by allowing students to migrate from public to private schools. In large-scale voucher systems, vouchers could also affect educational outcomes through a resorting of students across schools and a potential increase in competitive pressure among schools. Evidence on small-scale voucher programs includes Wolf et al. (2010); Peterson et al. (2003); Mayer et al. (2002); Chingos and Peterson (2015); Rouse (1998); Witte et al. (2012); Abdulkadiroglu, Pathak, and Walters (2015); Angrist et al. (2002); and Angrist, Bettinger, and Kremer (2006). Evidence on large-scale

on educational outcomes. Progressive vouchers could increase educational outcomes for low-income students by allowing them to migrate to higher-performing schools. Also, if schools are ordered to spend the extra resources from progressive vouchers on eligible students (as was the case in Chile), that could lead to more resources being devoted to them. When a program's scale is large, progressive vouchers could also affect educational outcomes through a re-sorting of students across schools and a potential increase in competitive pressure.

This study suggests that progressive vouchers do not lead to a change in students' distribution across schools. I also find that educational outcomes do not improve in the eligibility margin, indicating that even in contexts in which schools are incentivized to invest the extra resources in low-income students, these incentives are unlikely to lead to better educational outcomes in the margin. This implies that small-scale programs are unlikely to lead to better educational outcomes for low-income students. Large-scale voucher programs, such as the Chilean one, could still lead to better educational outcomes if they promote greater competition among schools, as suggested by Neilson (2013).⁴ However, if that is the case, the benefits are likely to be captured by all students and not just those who receive the extra voucher amount.

The work also contributes more specifically to the existing literature that looks at the effects of the Chilean reform. From 2008 to 2012 there was an increase in test scores for low-income students that has been attributed by many, with the exception of Feigenberg, Rivkin, and Yan (2019), to the reform (MINEDUC 2012; Correa, Parro, and Reyes 2014; Villarreal 2012; Mizala and Torche 2013; Neilson 2013; Navarro-Palau 2017). However, the 2008 reform brought a number of changes that could have led to better outcomes. Results could be due to increased resources, increased accountability that came attached to those resources, low-income parents' being able to choose better schools, or schools competing to enroll low-income students. More importantly, it is unclear whether progressive vouchers play a role in explaining these positive effects. If results are driven by increased resources or increased accountability, as suggested by Murnane et al. (2017) and Correa, Parro, and Reyes (2014), educational improvements could have been achieved without progressive vouchers. This paper provides further evidence on the mechanisms that could be at work. In particular, I am able to show robust evidence that results are not driven by low-income parents' being able to choose better-performing schools. If educational improvements are a result of progressive vouchers, as suggested by Neilson (2013), then results would have to be driven by increased competition for low-income students.

Results also relate more broadly to the existing literature that looks at barriers preventing individuals from making what could be optimal educational choices. This work shows that it is not just income and price that restrain low-income parents from choosing better-performing schools. Other barriers could include behavioral barriers, such as lack of information or the complexity associated with evaluating a substantial number of options, which previous research has found can drive students to make suboptimal

voucher programs includes Muralidharan and Sundaraman (2015); Figlio and Rouse (2006); Gallego (2013); Chakrabarti (2008); Rouse et al. (2013); and Figlio and Hart (2014). See Epple, Romano, and Urquiola (2015) for a more extensive review of this literature.

4. In a recent paper Andrabi, Das, and Khwaja (2017) also suggest that in educational markets, changes in consumer behavior can have important effects on the equilibrium level of school quality.

choices at the tertiary level (Thaler and Mullainathan 2008; Radford 2013; Pallais 2015; Smith, Hurwitz, and Howell 2015; Ross et al. 2013). Issues of social belonging could also play a role. Previous research has found that high-achieving low-income students tend to choose colleges that mimic the choices of their socioeconomically similar peers, despite their being more academically advantaged (Hoxby and Avery 2013). Also, at the secondary level, there is evidence that parents of each race prefer schools in which their own race is the clear majority, implying that minority parents face much larger trade-offs between academics and social preferences when choosing schools (Hastings, Kane, and Staiger 2009). Other barriers could include supply-side barriers, such as school screening. However, this likely plays a minor role in this setting in which schools were given strong incentives to admit eligible students.⁵

Results are also highly relevant from a public policy perspective. As school vouchers continue to be implemented across the world, and as the United States moves toward increasing school choice, we can learn much from the Chilean voucher experience. Results are also of course particularly relevant for the Chilean context, where a number of voucher system reforms are now being implemented.

II. Institutional Background

A. Voucher System

Chile has had a universal voucher system since 1981. The system has three types of schools: (i) public schools that are managed by local municipalities and that represent approximately 38 percent of total enrollment, (ii) private voucher schools that represent approximately 54 percent of total enrollment, and (iii) unsubsidized private schools, intended to attract upper-income households, that represent approximately 8 percent of total enrollment. Until 2008, the government paid both public and private voucher schools a flat voucher rate per student based on attendance.

The Chilean voucher scheme is unique in that it imposes few restrictions on private voucher schools. These schools can receive voucher subsidies regardless of their religious status, can operate for profit, can implement admission policies subject to few restrictions, and as of 1994, can also charge add-ons to parents. These add-ons are capped at about three times the voucher payment amount, but this constraint is rarely binding.⁶ Public schools face more restrictions; they are not allowed to charge add-ons to parents and cannot turn away students unless they are overenrolled.

Based on the particularities of the Chilean voucher design, theoretical models such as Epple and Romano (1998) and MacLeod and Urquiola (2012, 2015) suggest that the

5. The program involved an additional per-student monthly voucher amount of more than \$65 in 2012 in a setting in which 60 percent of private voucher schools that joined the program did not charge add-ons to eligible students, and the 40 percent that did, charged a median and mean monthly add-on of \$35 and \$41. This should be a strong incentive to admit eligible students, particularly those in the sample who have the highest SES among eligible students. The fact that I do not observe positive effects for students in the sample with more educated mothers also indicates that results were nonsignificant even among students who should have faced fewer supply-side barriers.

6. According to Urquiola and Verhoogen (2009), most elite unsubsidized private schools could take vouchers but choose not to accept them.

voucher system would lead to cream-skimming from the public sector and stratification by income and/or ability within the private sector, which is exactly what is observed in practice.⁷

Although it is hard to isolate the causal effect of the voucher system on student sorting, existing research suggests that the voucher system has led to increased socioeconomic stratification across schools. Among the relevant research, Hsieh and Urquiola (2003, 2006) implement a difference-in-difference strategy in which they compare stratification measures across municipalities with more or less growth in the private sector and find that the voucher-induced growth in the private sector is associated with a middle-class exodus from public schools. In a related work, McEwan, Urquiola, and Vegas (2008) compare student sorting across towns of different population sizes. On the basis of the idea that there is a minimum population size required for private school entry and that towns close to this threshold are comparable, the authors find that private school entry is related to higher student stratification.

There is a broader cross-sectional body of literature in Chile that documents the high levels of stratification by SES. Valenzuela, Bellei, and De los Ríos (2010) suggest that Chile has one of the highest levels of school-level stratification by SES in the OECD. Moreover, Mizala, Romaguera, and Urquiola (2007) show that stratification is particularly extensive in private-sector schools.⁸

B. Progressive Vouchers

The existing research on the high levels of segregation by SES in the Chilean system was in part what led to the 2008 reforms. The reform, *Ley de Subvención Escolar Preferencial*, increased the voucher amount for students in the lowest 40 percent of the income distribution. Figure 1 shows how voucher amounts evolved from 2006 to 2012 for eligible and noneligible students. As can be seen in the figure, in 2012, eligible students were receiving approximately \$65 extra per month, which represented an increase of roughly 50 percent over the regular voucher amount.

The Ministry of Education determined eligibility for the extra voucher amount using administrative records. Therefore, parents did not have to actively apply. Students were considered to be low-income if: (i) they belonged to the lowest 30 percent of the income distribution according to the socioeconomic ranking, which is designed to measure the risk of being in poverty; (ii) they belonged to the social program *Chile Solidario*, which is a component of the Social Protection System in charge of serving families, individuals, and areas in socially vulnerable conditions; or (iii) they met certain other poverty requirements. Once students were classified as eligible, the information was posted online, and individuals and schools could easily check whether a specific student was eligible for progressive vouchers before making any enrollment decision.

The extra resources were made available to all public schools as well as private voucher schools that signed up for the program. Schools that chose not to participate in

7. In the theoretical model developed by Epplé and Romano (1998), stratification occurs as a consequence of positive peer effects. Instead, in the model developed by MacLeod and Urquiola (2012, 2015), stratification occurs because employers use an individual's school of origin as a signal of their skill.

8. See Epplé, Romano, and Urquiola (2015) for a more extensive review of the Chilean voucher system.

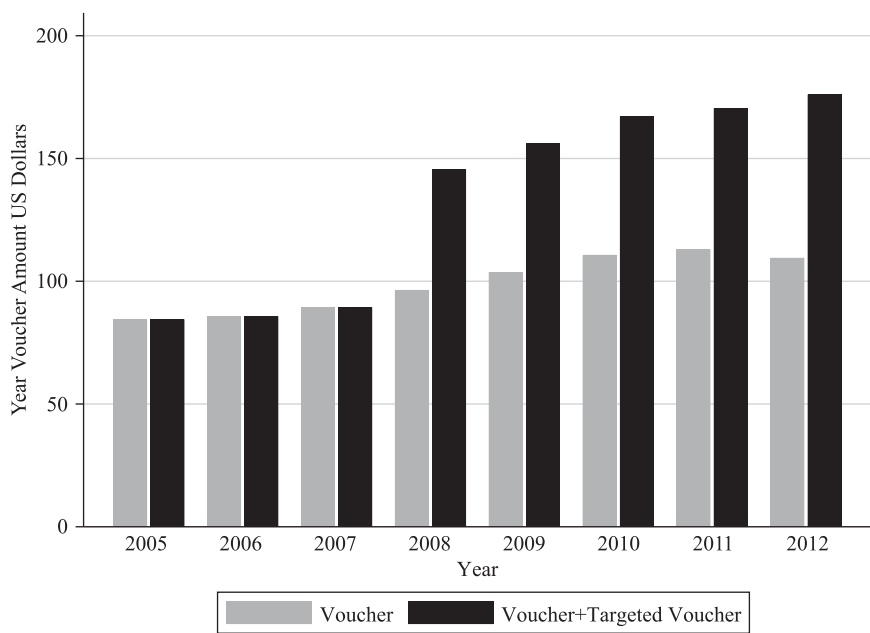


Figure 1
Voucher Amounts over Time

Notes: The figure shows how the voucher amount evolved over time for eligible and noneligible students. All amounts are in 2012 U.S. dollars and represent a one-month payment. The voucher presented is for students in first grade at schools with full sessions.

the program could still receive the regular voucher amount for each student but could not receive the extra resources from progressive vouchers. In the following I describe the type of schools that chose to join the program. This analysis is key to better understand how the policy changed choice sets for eligible parents. I then provide information on how schools might have spent these extra resources. This second analysis is relevant to understanding how the policy might—or might not—have impacted educational outcomes. Finally, I describe how test scores evolved during this period and the existing evidence regarding the effects of the 2008 reform.

1. Schools that signed up for the policy

Schools that signed up to receive the extra voucher amount had to agree to: (i) provide a detailed accounting of how the progressive voucher funds were used—something that is not required with the regular voucher; (ii) present an educational improvement plan to the Ministry of Education that included detailed education reforms the school would undertake to improve test scores; (iii) define anticipated test score gains for future years, particularly for eligible students; and (iv) not charge add-ons to eligible parents, although they could still charge add-ons to noneligible parents.

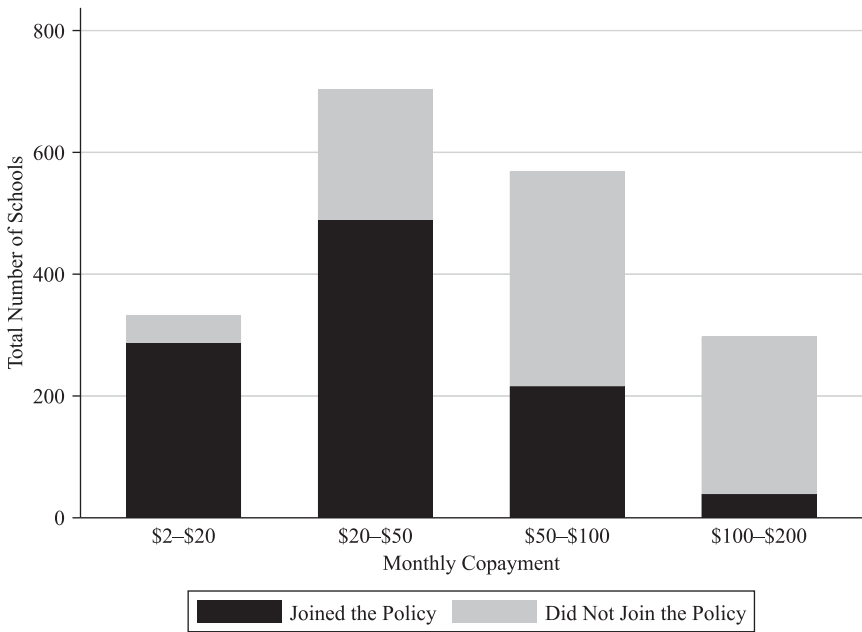


Figure 2

Private Voucher Schools That Had and Had Not Joined the Program in 2012 by Monthly Add-Ons Charged to Parents

Notes: The figure shows the number of private voucher schools serving primary education that had and had not joined the program in 2012 depending on the monthly add-ons they charged parents.

Schools were also required to eliminate screening of eligible students based on past academic performance and family background. However, this was not enforced in practice. Carrasco et al. (2014) provide qualitative evidence that, until recently, screening in high-demand schools, both those schools that joined the program and those that did not, was openly practiced. Common practices included “game sessions” for first-grade students aimed at measuring skills that could predict good behavior, development, and adaptation, as well as interviews with prospective parents.⁹

The new program, therefore, offered increased revenues for schools, but came at a cost. In practice, by 2012, all public schools had joined the program, and approximately 95 percent of private vouchers schools that charged no add-ons to parents had joined. However, among private voucher schools that charged add-ons to parents, only about 50 percent had joined. Figure 2 shows the number of private voucher schools that were charging monthly copayments between \$2 and \$200 in 2012. Black bars indicate the

9. The government has since implemented a centralized system of school admission to make sure that over-subscribed schools run a lottery to select their students.

Table 1
Schools' Characteristics

	Enrollment (1)	Test Scores (2)	SES (3)	Add-On (4)	Size (5)	Class Size (6)	Capacity Constrained (7)
Joined the program							
Public	57%	249.8 (27.7)	9.0 (1.8)	0.0 (0.0)	23.3 (25.7)	16.5 (12.6)	0.07 (0.25)
Private voucher w/o add-on	19%	247.4 (28.1)	9.1 (2.2)	0.0 (0.0)	25.1 (28.3)	18.0 (14.2)	0.08 (0.28)
Private voucher w/add-on	13%	261.6 (21.3)	11.6 (1.1)	41.0 (31.3)	53.4 (38.5)	31.6 (9.9)	0.07 (0.25)
Did not join the program							
Private voucher w/add-on	11%	272.9 (19.6)	12.9 (1.1)	86.0 (47.0)	55.7 (39.5)	30.5 (9.5)	0.08 (0.27)

Notes: Table 1 presents average characteristics in 2012 for different types of subsidized schools. Test score equals the average result on the Grade 4 standardized test in 2011, SES equals the average years of education of mothers' of students attending those schools, add-on equals the total amount charged to noneligible parents in those schools (USD), school size equals the cohort size at those schools, class size equals the average class size at those school, and capacity constrained equals the percentage of schools that surpass the legal class-size cap.

number of schools in each category that had joined the program by 2012, and gray bars show the number of schools in each category that had chosen not to join the program. As shown, the majority of schools that were charging between \$2 and \$50 had chosen to join the program, but few schools with monthly add-ons more than \$50 had joined.

Table 1 presents average characteristics for public schools, private voucher schools that charge no add-ons, and private voucher schools that charge add-ons and had chosen either to join or not to join the program. All measures are for 2011, the year when parents in my sample made their choices.¹⁰ The table shows that private voucher schools that charged high add-ons and/or served high-SES students abstained from participating in the program. Still, Table 1 also shows that the policy gave low-income students free access to a subgroup of schools that in 2011 had higher test scores (their average academic performance was in the 66th percentile) and had higher SES students (the average SES status of their students was in the 78th percentile) than public schools and private voucher schools that charged no add-ons. Table 1 also indicates that only about 7

10. Because the policy was implemented in 2008, these measures may already reflect some of the changes induced by the policy. [Online Appendix 1](#) presents the average characteristics for these schools in 2007, the year before the policy was implemented. Data show that from 2007 to 2011, public schools and private voucher schools that had joined the program increased their test scores. Still, it is always the case that private voucher schools with add-ons that joined the program are better than public schools and private voucher schools that charge no add-ons to parents.

percent of schools surpassed the legal class-size cap of 45, suggesting that capacity constraints were not much of an issue.¹¹

2. Use of resources from progressive vouchers

To join the program, schools had to agree, among other things, to present an Educational Improvement Plan (EIP) detailing the educational reforms that the school would undertake to improve academic results. The plan was also meant to set academic goals, particularly for eligible students. The EIPs contained specific actions that schools would take in areas such as curriculum management, school leadership, student life, and resource management. Resources from progressive vouchers could be spent on, among other things, hiring teachers, educational assistants, and the other necessary staff to meet the goals of the EIP. However, schools were not allowed to use these resources for increased salaries, bonuses, or other expenditures, such as debt repayment or school celebrations (Feigenberg, Rivkin, and Yan 2019).

The Ministry of Education classified schools in three ranked categories according to their academic performance and students' SES. Schools in higher categories had more freedom to define their EIPs and decide how to allocate the progressive voucher resources. In contrast, schools in the lowest category had to elaborate their EIPs with the assistance of the Ministry of Education. Sanctions for noncompliance with the EIP depended on the school's classification. Schools in higher categories could be demoted to a lower category, and schools in the lowest category could eventually be closed. In practice, in 2008 when the policy was first implemented, no school was assigned to the lowest category. By 2012, roughly 3 percent of schools had fallen to the lowest category, but none of them had been closed.

Even though the reform was meant to hold schools accountable for the extra resources from progressive vouchers, in practice, a number of schools did not comply with the requirements, a point emphasized by Feigenberg, Rivkin, and Yan (2019). Information in that paper comes from an audit conducted by the Chilean Comptroller's Office in 2012 that compared the funding inflows from progressive vouchers for 2008–2011 to documented expenditures in 77 of 345 municipalities. According to Feigenberg, Rivkin, and Yan (2019), on average, only 65 percent of received funds could be linked to validated expenditures.

3. Existing evidence on the effects of progressive vouchers

Stylized facts for this period indicate that, after years of stagnation, state-mandated test scores increased in almost 0.3 standard deviations from 2007 to 2012. This increase in overall test scores was driven mainly by an increase in the academic results of low-SES students. Differences in test scores between students with SES in the bottom 40 percent and those with SES in the top 60 percent, as measured by their mothers' education, diminished roughly from 0.69 standard deviations in 2005 to 0.49 standard deviations in 2012. Differences are less pronounced when we compare eligible to noneligible

11. It is difficult to know how many schools were oversubscribed in 2012, as I cannot determine whether the number of applicants surpassed the number of vacancies in each school. I can assume that schools with class sizes at or above 45 are capacity-constrained, as this is the legal class-size cap.

students, that is, students with SES in the bottom 40 percent and the top 60 percent as defined by socioeconomic rankings. Still, differences between eligible and noneligible students declined from 0.37 standard deviations in 2007 to 0.3 standard deviations in 2012 (see Appendix 1).

Many people have attributed this increase in test scores to the 2008 reform, which is a conclusion derived primarily from the common findings that schools that chose to join the program and that should have been most affected by it saw significant increases in test scores and that test scores significantly increased for low-income students who should have been most affected by the program. Existing studies include MINEDUC (2012); Correa, Parro, and Reyes (2014); Villarroel (2012); Mizala and Torche (2013); Murnane et al. (2017); Neilson (2013); and Navarro-Palau (2017).¹² A more recent paper by Feigenberg, Rivkin, and Yan (2019) does not concur with these findings; they instead find that changes in parental education and household income could account for much of the reduction in the achievement differential between low- and high-SES students observed in the data.

Although most studies conclude that the program improved educational outcomes for low-income students, little is known about the potential mechanisms that could be driving the effects. The 2008 reform brought numerous changes that could have led to better educational outcomes for low-income students. Effects could be a result of the increased revenues received for each eligible student by public schools and private voucher schools that chose to join the program. Effects could be a consequence of schools' increased accountability under the program. Results could also be a direct effect of increased competition between schools for low-income students, or they could be a consequence of low-income students' being able to choose better schools given the expansion of their choices.

Not only is there little evidence regarding the mechanisms that could be at work, but there is also uncertainty regarding whether implementing a differentiated voucher was at all important in terms of explaining the improvements. If results are driven by increased revenues or increased accountability, as suggested by Correa, Parro, and Reyes (2014) and Murnane et al. (2017), then, in principle, it could have been possible to increase test scores for low-income students without the use of a progressive voucher. If instead, results are driven by increased competition or by low-income parents' being able to choose better schools, as suggested by Neilson (2013), then having a differentiated voucher system would be key in explaining previous results.

Existing studies typically conclude that given the magnitude of the effects, it seems unlikely that educational improvements are driven solely by a reallocation of students across schools. However, there is much debate regarding the extent to which parents changed their choices for their children as a result of the reform. Looking at overall trends in enrollment, Murnane et al. (2017) concludes that the reform led to a major

12. MINEDUC (2012); Correa, Parro, and Reyes (2014); Villarroel (2012); Mizala and Torche (2013); and Murnane et al. (2017) all compare the academic outcomes of schools that chose to join the program with the academic outcomes of schools that chose not to join and find a positive impact on test scores that ranges between 0.08 and 0.2 standard deviations. Neilson (2013) compares the academic outcomes of low- and high-income students and finds that progressive vouchers raised test scores for low-income students by 0.2 standard deviations. Navarro-Palau (2017) uses variation from date of birth enrollment cutoffs to compare the outcomes of students who had different exposures to the reform and finds significant, though more modest, effects of the reform on school achievement.

increase in enrollment rates at private voucher schools charging add-ons. Navarro-Palau (2017) instead uses a different method, in which she exploits variation from the date of birth enrollment cutoff and compares the choices of students who were more or less exposed to the program, and concludes that only a small fraction of students changed their enrollment decisions as a result of the reform. However, a major drawback in both studies is that they require important assumptions about trends in enrollment rates. Moreover, because they cannot determine which students in previous cohorts would have qualified for the extra voucher amount, their analyses rely on overall enrollment rates. This of course could be masking effects if there are, for example, capacity constraints. On a different note, Neilson (2013) uses a structural model of school demand and supply and constructs counterfactual simulations to isolate the different mechanisms through which the policy could have affected outcomes. The author concludes that approximately one-third of the observed improvement could be due to eligible families being able to choose better schools because of the larger voucher amount, and two-thirds of it could be due to the rise in the quality of existing schools in response to increased competition. However, as is always the case with these types of models, it involves making strong assumptions about the demand and supply of schools.

This work contributes by providing robust evidence on the effect that having access to progressive vouchers had on students' school choices and educational outcomes. Results help to document the extent to which progressive vouchers helped to reallocate students across schools and mitigate socioeconomic stratification, as well as determine the extent to which parents' reassignment choices could have led to better educational outcomes. Although local in nature, results indicate that increases in test scores were not driven by low-income students migrating to better schools. More evidence is needed to determine whether improvements throughout this period were driven by increased resources, increased accountability, increased school competition, or something else.

III. Data and Descriptive Statistics

In this study, I use a unique national data set containing detailed information at the individual level. What makes the data unique is information on students' socioeconomic ranking (Ficha de Proteccion Social) for 2012, which is the variable used to determine program eligibility. This variable was provided by the Ministry of Education and the Ministry of Social Development and was not available for previous studies.

I merged information on program eligibility with administrative records on school enrollment that cover the whole population of students. To characterize schools, I use administrative data on schools' locations, prices, type of holder (that is, private versus public), overall size, and class size. Data from the SIMCE, a yearly, nationwide test for students in Grades 2, 4, 6, and 8, is used to measure schools' academic achievement and students' educational outcomes.

To characterize students, I use information from questionnaires given to parents of students in Grades 2, 4, 6, and 8, together with the SIMCE. These questionnaires allow me to gather information about the educations of the students' mother and father, the number of books in the house, monthly income, and childcare attendance.

This information complements detailed enrollment records that document the students' age, gender, and exact location. Data on students' exact addresses come from enrollment records, where schools report parents' addresses. This information is only available for about 40 percent of students in the sample.¹³

For simplicity, I restrict my analysis to students entering first grade in 2012. The choice to focus on first-grade students has to do with analyzing students who are relatively free to choose any school and who face no costs associated with changing schools. Still, results remain the same when looking at students in Grades 1–8.¹⁴ Because the socioeconomic ranking changes from year to year, and I only have good quality data on this variable for 2012, I restrict my analysis to 2012—four years after the policy was first implemented. The benefit of focusing on this period is that we expect that parents and students by then must have had a good sense of how the policy actually worked. Still, I am able to show that results remain roughly the same when focusing on 2008 or 2009.

IV. Empirical Strategy

To estimate the causal effect of the differentiated voucher, I exploit the fact that the policies' eligibility process generates large discontinuities in the relation between the student's socioeconomic ranking and the probability of being eligible. Although there are other criteria that can determine eligibility, the socioeconomic ranking accounts for the admission of about 83 percent of all participants in the program (that is, it is the binding threshold).¹⁵ This allows me to implement a fuzzy RDD. The estimating equations can be written as:

$$(1) \quad y_i = \alpha_0 + \rho D_i + f(r_i) + \delta'X + \epsilon_i$$

$$(2) \quad D_i = \gamma_0 + \pi 1[r_i \leq r_0] + g(r_i) + v'X + \mu_i$$

where y_i is the variable of interest, for example, academic performance of student i ; D_i equals one if the student is eligible for a progressive voucher; r_i is the running variable (student's socioeconomic ranking); $1[r_i \leq r_0]$ is an indicator function that equals one if the student is below the threshold for program eligibility; X are students' socioeconomic characteristics; and ϵ_i and μ_i are error terms. Throughout the paper I use a bandwidth of 1,200 points. This corresponds closely to the optimal bandwidth suggested by Calonico, Cattaneo, and Titiunik (2014). Such optimal bandwidth varies by outcome. However, I choose to fix the bandwidths across regressions to have a single sample in each case. Estimates remain the same when using alternative bandwidths (see [Online Appendix 3](#)).

13. [Online Appendix 2](#) shows that students with address information have slightly higher SES characteristics, but are not substantially different from the general sample used in the study.

14. Results available upon request.

15. As mentioned in Section II.B, students can also qualify if: (i) they belong to the social program Chile Solidario, which is a component of the Social Protection System in charge of serving families, individuals, and areas in socially vulnerable conditions or (ii) they meet certain poverty requirements. However, under the second case students will only be beneficiaries for a year and will have to be reevaluated under the government's socioeconomic ranking during the next year in order to retain the benefit.

The socioeconomic ranking used to determine eligibility is an instrument designed to measure the risk of being in poverty. It considers a household's ability to generate income based on education, experience, and county of residence and a household's economic need based on, among other things, the number of children. The socioeconomic ranking is used to determine eligibility for a number of other welfare programs and was created in 2007, prior to the introduction of progressive vouchers.¹⁶ This ranking can change from year to year, either because the ages of household members change or because households choose to be reevaluated in order to apply to other welfare programs, which typically requires up-to-date information. If a student falls above the cutoff on a given year, they will no longer be eligible for progressive vouchers. Importantly, the threshold for progressive vouchers does not overlap with the threshold for any other program.

V. Results

A. First-Stage Estimates

I begin by showing how crossing the socioeconomic threshold for program acceptance affects the probability that a student becomes eligible for the targeted voucher, as well as their probability of actually receiving the extra voucher amount. In practice, not all students who are eligible for targeted vouchers receive the extra voucher amount, as some of them choose to enroll in private schools or private voucher schools that did not join the program.

Results can be found in Figures 3A and 3B. In each case, black lines depict a fourth-order polynomial fit for control and treatment units separately, and gray dots represent the sample average for each disjoint bin. As can be seen, being below the threshold for program acceptance increases the probability that a student becomes eligible for a targeted voucher by 70 percentage points, and it increases the probability that a student receives the extra voucher amount by 60 percentage points.

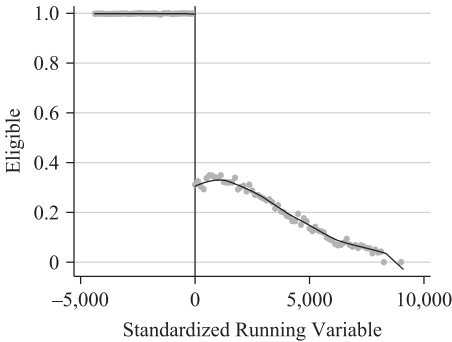
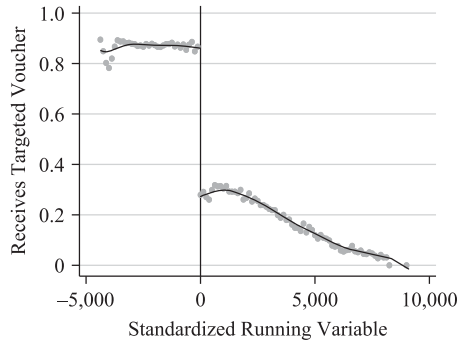
Figure 3 shows that students below the cutoff are always eligible. However, the discontinuity is not sharp because students above the cutoff can still qualify for progressive vouchers if: (i) they belong to the social program Chile Solidario, which is a component of the Social Protection System in charge of serving families, individuals, and areas in socially vulnerable conditions, or (ii) they meet certain poverty requirements.¹⁷ All students who belong to the Chile Solidario program are automatically eligible; however, parents who meet certain poverty requirements need to actively apply in order for their children to become eligible.

Using Abadie (2002), I look at the characteristics of compliers¹⁸ in the sample and find that compliers are not very different from the general population of students who are

16. Before 2007 the government used a different instrument, the Ficha CAS, to evaluate household SES.

17. Students can become eligible if they belong to the lowest category in the public health system. Alternatively, other variables are considered, such as income, education, and the location of the household.

18. Complifiers are students who become eligible for the extra voucher amount because their socioeconomic ranking is below the threshold for program eligibility, but who would not be eligible for the program otherwise.

Panel A: Eligible for Progressive Voucher**Panel B: Receives Progressive Voucher****Figure 3***First Stage*

Notes: This figure shows how the probability of being eligible for the extra voucher amount and the probability of receiving the extra voucher amount both jump at the threshold for program acceptance. Gray dots present the average of the outcome variable for individuals in equally spaced disjoint bins. Black lines depict a fourth-order polynomial fit for control and treatment units separately.

close to the cutoff. However, they have slightly higher socioeconomic characteristics, indicating that always-takers¹⁹ in the sample come from a lower SES (see [Online Appendix 4](#)).

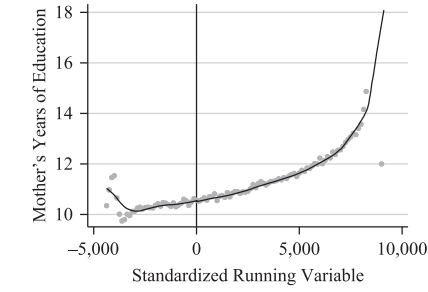
Among always-takers, approximately 40 percent are eligible because they belong to Chile Solidario, and 60 percent are eligible because they meet certain poverty requirements. There is a slight difference between both groups of always-takers. Although all students who belong to Chile Solidario continue to be eligible in the following year (as are students who meet the socioeconomic ranking requirements), only 70 percent of students who meet certain poverty requirements continue to be eligible in the following year. This happens because the law requires these students to be reevaluated by their socioeconomic ranking during the following year, in order not to lose their benefit. What this implies for the analysis is that roughly 60 percent of always-takers are receiving what could be interpreted as a lower treatment dose because they are at greater risk of losing their benefit in the future. If anything, this could magnify instrumental variable estimates. True instrumental variable estimates would be closer to reduced-form effects. In practice, this does not change the interpretation of results.

B. Balance Checks

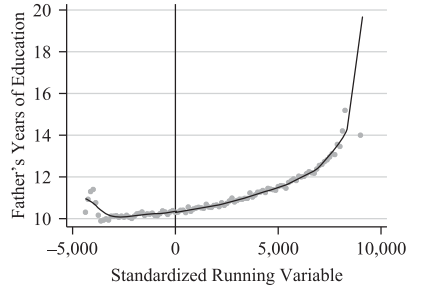
Next, I perform standard balance checks to examine whether individuals just above and just below the socioeconomic cutoff are similar in terms of their observable characteristics. I look at a set of socioeconomic variables that should not be affected by the program. If the procedure is valid, then RDD estimates should be equal to zero.

19. Always-takers are students who become eligible for the extra voucher amount regardless of their socioeconomic ranking and who would be eligible for the program even if they were above the socioeconomic threshold.

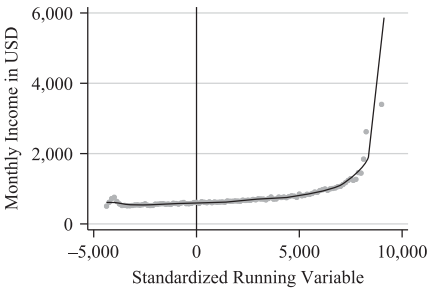
Panel A: Mother's Years of Education



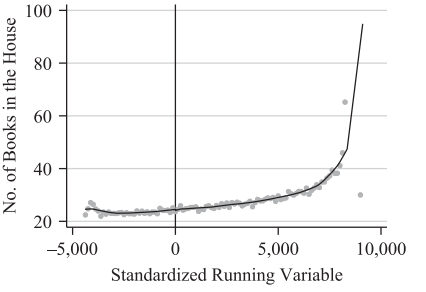
Panel B: Father's Years of Education



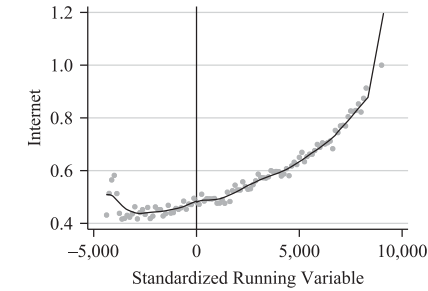
Panel C: Monthly Income



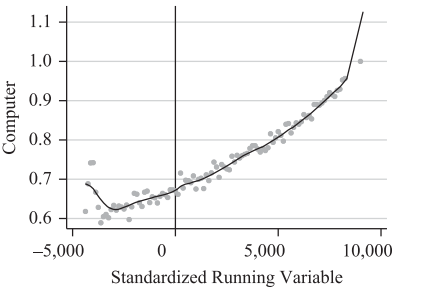
Panel D: Books in the House



Panel E: Internet



Panel F: Computer



Panel G: Attended Childcare 0–2 Years

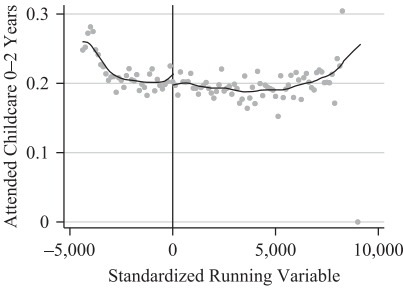


Figure 4
Visual Evaluation of Robustness Checks I

Notes: This figure shows that students' socioeconomic characteristics change smoothly across the threshold. Gray dots present the average in the outcome variable for individuals in equally spaced disjoint bins. Black lines depict a fourth-order polynomial fit for control and treatment units separately.

Table 2
Robustness Check

	Mother's Education (1)	Father's Education (2)	Monthly Income USD (3)	Books (4)	Internet (5)	Computer (6)	Attended Childcare (0–2) (7)
$R \leq \text{cutoff}$	0.045 (0.083)	0.060 (0.088)	−10.562 (14.725)	0.344 (0.687)	0.003 (0.015)	−0.001 (0.014)	0.018 (0.012)
Mean control	10.538	10.286	604.059	24.427	0.488	0.676	0.199
Observations	22,941	21,998	23,015	22,972	20,448	22,366	22,865

Notes: Table 2 shows that students around the cutoff are similar in terms of their socioeconomic characteristics. Results from *rdrobust* (Calonico, Cattaneo, and Titiunik 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 4 and Table 2 present the results. *Mother's education* and *father's education* include total years of education. *Income* equals monthly income (USD). *Books* equal total number of books in the house. *Internet* and *computer* are both dummies that equal one if the student had internet or a computer in their house. *Attended childcare* is a dummy variable that equals one if the student attended childcare from zero to two years old.

Figure 4 displays the binned mean of observable characteristics of students by socioeconomic score relative to the cutoff. As can be seen, all of the studied characteristics change smoothly across the threshold. Regression estimates in Table 2 confirm the visual analysis; results are small in magnitude and precisely estimated, indicating that students around the cutoff are similar in terms of their socioeconomic characteristics.²⁰

To further check whether there is any sign of socioeconomic score manipulation, I look at whether there is any evidence of a visible jump in the density around the discontinuity. Figure 5A shows a histogram of scores relative to the admission cutoff value, and Figure 5B shows the result from the McCrary (2008) test. As can be seen, there is no visible jump in the density around the discontinuity. The McCrary (2008) test confirms this, showing no evidence of a statistically significant break.²¹

The fact that I do not observe any evidence of score manipulation near the cutoff is to be expected. Eligibility for progressive vouchers is determined by the Ministry of Education on the basis of administrative data and does not require an application on the part of parents. Also, information on cutoff scores is not made available to parents. Although there is some anecdotal evidence of socioeconomic ranking manipulation, this tends to occur at other cutoff points, where other welfare programs, such as housing, are assigned.

20. Socioeconomic variables are available for approximately 70 percent of students in my sample. Missing rates are also smooth around the cutoff.

21. The log difference in height is 0.014 with a standard error of 0.020.

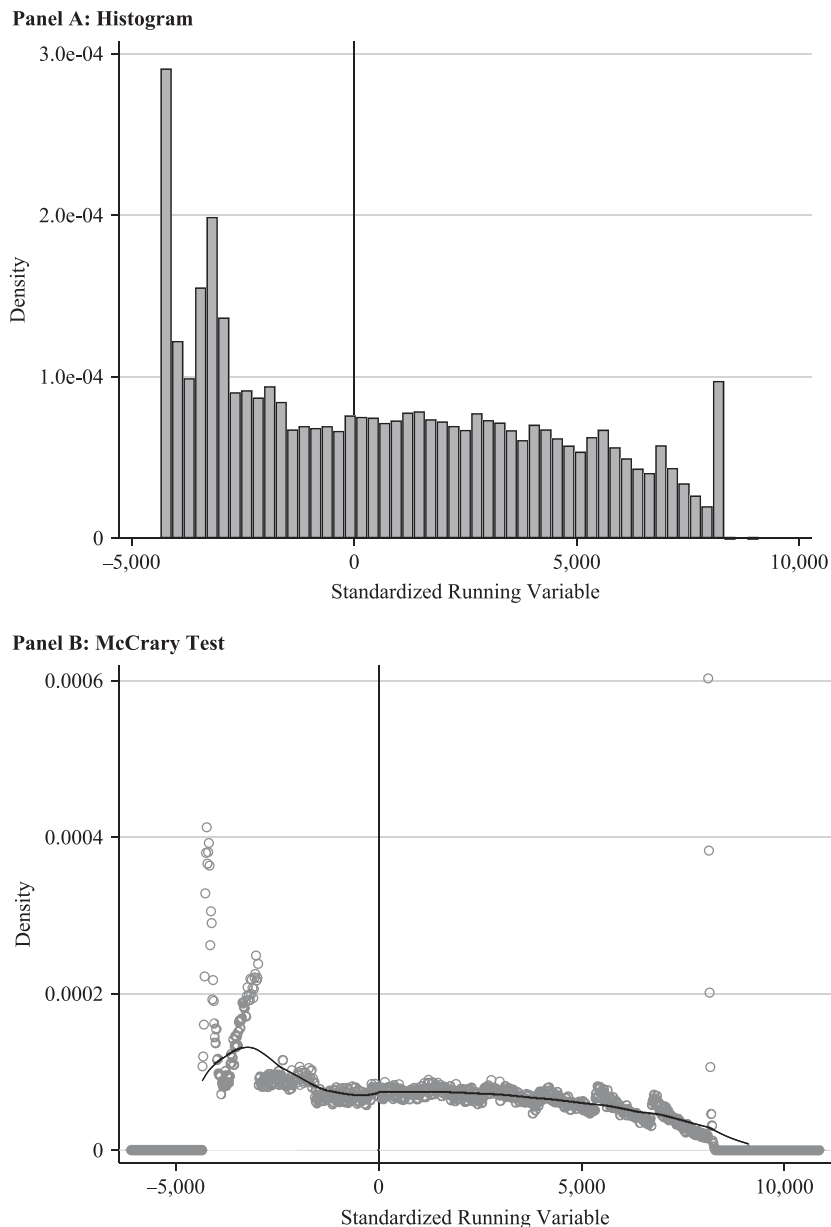


Figure 5
Visual Evaluation of Robustness Checks II

Notes: This figure shows that there is no evidence of socioeconomic manipulation across the threshold. Panel A shows a histogram of scores relative to the cutoff value, and Panel B shows the results of the McCrary (2008) test, $r(\text{bandwidth}) = 1742.61$, $r(\text{binsize}) = 17.373$, $r(\text{se}) = 0.020256$, $r(\text{theta}) = 0.01413$.

C. *Progressive Vouchers and School Choice*

I now turn to estimating the impact of eligibility for progressive vouchers on school choice. I expect that being eligible for progressive vouchers will potentially expand school choice sets for parents and change the relative prices that parents have to pay for school. Eligibility should expand choices because schools that join the program should now be more willing to accept eligible students (because they receive more money) as opposed to noneligible students. Eligibility should also change relative prices because eligible students now do not have to pay add-ons to attend schools in the program.

The impact that progressive vouchers might have on enrollment decisions depends on parents' preferences. In general, previous studies have found that when it comes to school choice, parents value proximity highly, and parental preferences for test scores increase with students' income and own academic ability (Bayer, Ferreira, and McMillan 2007; Hastings and Weinstein 2008). Similar results have been found for Chile; authors have documented that quality is a superior attribute and closeness to home an inferior attribute, suggesting that poor families tend to place more value on closeness to home rather than school quality (Gallego and Hernando 2008; Chumacero, Gómez, and Parades 2011).

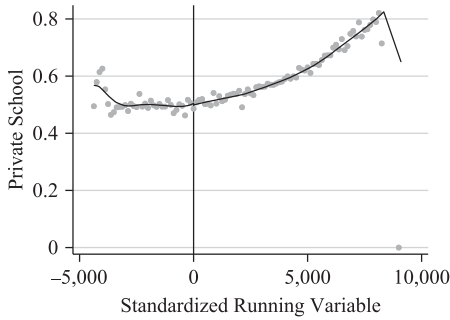
In general, I expect that being eligible for progressive vouchers should lead parents to choose schools with higher test scores or higher SES. However, because not all schools chose to join the program, it is possible that the change in relative prices leads parents to choose schools with lower test scores or lower SES. This would be the case if, had they not been eligible, parents would have chosen a private school or a private voucher school that did not join the program. As a point of reference, in 2007, the year before the policy was implemented, approximately 57 percent of students entering first grade who met the eligibility requirement for progressive vouchers enrolled in a public school, 16 percent enrolled in a private voucher school that charged no add-ons to parents, 18 percent enrolled in a private voucher school that charged add-ons to parents and that later joined the program, and only 9 percent enrolled in a private or private voucher school that did not join the program.²² Therefore, I do not expect to see much of a negative effect in terms of having parents choose schools with lower test scores or lower SES.

Figure 6 and Table 3 present the results. Chosen schools are characterized based on their observable characteristics in 2011, which is the year before parents made their choices. *Private* is a dummy variable that equals one if the chosen school is private. *Test score* equals the performance of the chosen school on fourth-grade standardized test scores. *SES* equals the average years of education of the mothers of students attending the chosen school. *Add-on* equals the monthly amount charged by the chosen school to noneligible parents (USD). *Class size* equals the average class size of the chosen school. *Distance* equals the distance traveled to school in miles.²³ Schools' test scores and SES are both standardized by the mean and standard deviation of the variables in 2005.

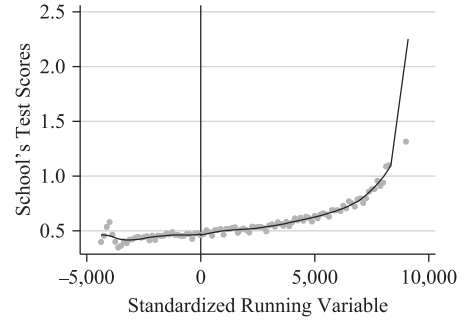
22. These numbers are similar for students who are close to the threshold.

23. I am only able to obtain the exact location for approximately 40 percent of students in my sample, which is why there are fewer observations in distance estimates.

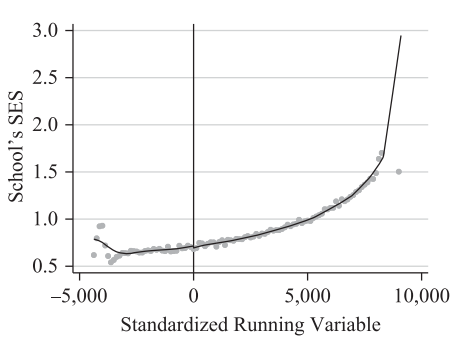
Panel A: Private School



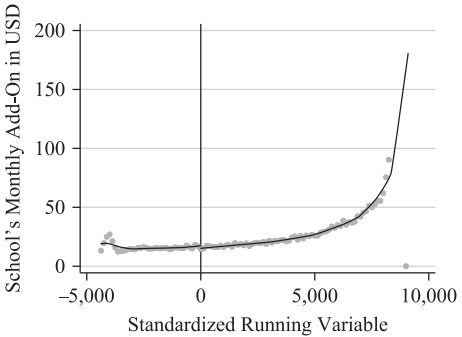
Panel B: School's Test Scores



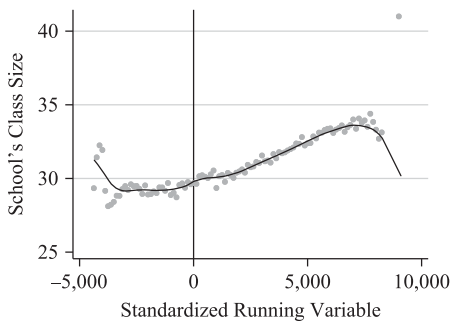
Panel C: School's SES



Panel D: School's Monthly Add-On



Panel E: School's Class Size



Panel F: Distance to School

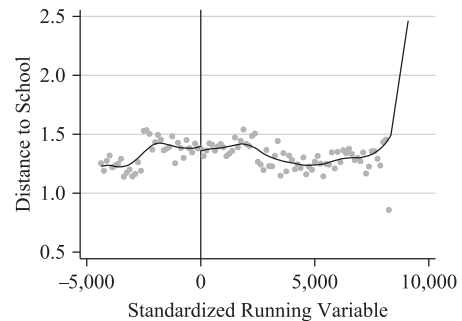


Figure 6
Visual Evaluation of Results

Notes: This figure shows results on school choice and educational outcomes. Gray dots present the average in the outcome variable for individuals in equally spaced disjoint bins. Black lines depict a fourth-order polynomial fit for control and treatment units separately.

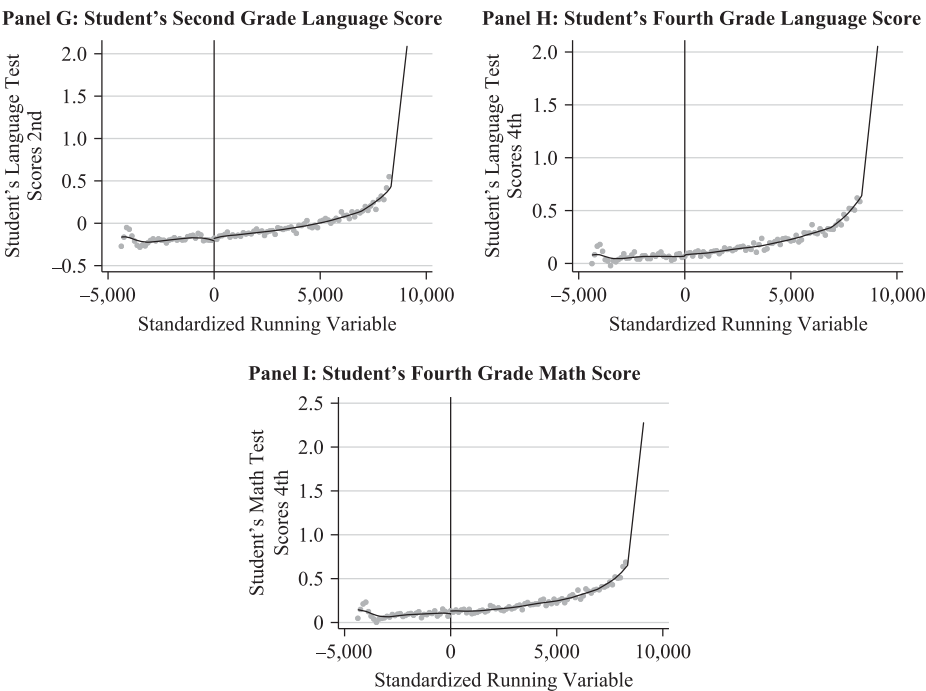


Figure 6 (continued)

Figures 6A–6F display the binned means of observable characteristics of the chosen schools by students’ socioeconomic score relative to the cutoff. As can be seen, all of the studied school characteristics change smoothly across the threshold. Estimates in Columns 1–6 of Table 3 confirm the visual analysis. Panel A contains reduced-form estimates, and Panel B contains instrumental variable estimates where the discontinuity is used as an instrument for being eligible for progressive vouchers. All estimates include controls for mother’s education, father’s education, household income, and region. Results remain the same when using alternative bandwidths (see [Online Appendix 3](#)). As can be seen, being below the cutoff has no impact on the probability of a student choosing a private school, the test scores of the chosen school, the SES of the chosen school, the average class size of the chosen school, or the distance traveled to school. Results do show that students below the cutoff choose schools that charge higher add-ons to noneligible parents. However, these results are small in magnitude, indicating that eligible students, on average, choose schools that charged \$3 more per month from an average monthly cost of \$15 for students in the control group.

D. Progressive Vouchers and Educational Outcomes

I next estimate the impact of being eligible for progressive vouchers on students’ educational outcomes. Although the program has no impact on parents’ school choices, it is still possible that the program has a positive impact on educational outcomes. First,

Table 3
School Choice and Educational Outcomes

School Private (1)	School Test Scores (2)	School SES (3)	School Add-On (4)	School Class Size (5)	School Distance (Miles) (6)	Student Language Second Grade (7)	Student Language Fourth Grade (8)	Student Math Fourth Grade (9)
Panel A: Impact of Being below the Cutoff (Reduced Form)								
<i>R</i> ≤ cutoff	0.004 (0.012)	0.006 (0.017)	2.049*** (0.746)	-0.016 (0.245)	0.039 (0.067)	-0.029 (0.027)	-0.008 (0.025)	-0.030 (0.023)
Mean control	0.498	0.470	15,346	29,883	1.371	-0.169	0.083	0.135
Observations	31,640	30,647	31,495	31,135	11,961	24,845	23,555	23,658
Panel B: Impact of Being Eligible for Targeted Vouchers (IV)								
<i>R</i> ≤ cutoff	0.005 (0.017)	0.008 (0.025)	2.948*** (1.075)	-0.023 (0.354)	0.053 (0.092)	-0.041 (0.038)	-0.012 (0.036)	-0.043 (0.033)
Mean control	0.498	0.473	14,491	29,966	1.362	-0.154	0.094	0.146
Observations	31,640	30,647	31,495	31,135	11,961	24,845	23,555	23,658

Notes: Table 3 shows effects on school choices and educational outcomes. All estimates include controls for mother's education, father's education, and region. Panel A contains reduced-form estimates, and Panel B contains instrumental variable estimates where the discontinuity is used as an instrument for progressive voucher eligibility. Results from rdrobust (Calonico, Cattaneo, and Titiunik 2014). Standard errors in parentheses. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

it is possible that the program allowed parents to choose schools that better meet their children's needs, although there are no significant differences in the observable characteristics of the chosen schools. Second, because schools received extra revenues as a result of the policy, they could have used these resources to improve the educational outcomes of eligible students.

I focus on students' educational outcomes in second and fourth grades. By fourth grade we observe that almost no student has dropped out of the system.²⁴ This is to be expected considering that these are relatively young students. However, roughly 30 percent of them have changed schools. Part of this is likely explained by students moving homes. In fact, the data show that in fourth grade, both eligible and noneligible students in the sample are attending schools that are on average very similar to the schools they had chosen in first grade, but that are further away from their original addresses. (Estimates can be found in [Online Appendix 5](#).)²⁵

Figures 6G–6I and Columns 7–9 in Table 3 present the results for educational outcomes. Figures 6G–6I display the binned means of test scores by students' socioeconomic score relative to the cutoff, and Columns 7–9 in Table 3 present estimates of students' performance in a nationwide language test given to students in second and fourth grades, and a nationwide math test given to students in fourth grade. Test scores for fourth grade are standardized by the mean and standard deviation of these variables in 2005. Test scores for second grade are standardized by the mean and standard deviation of the variable in that same year because there is no measure of second grade test scores for previous years. Results show no impact of the program on students' educational outcomes in the short or medium terms. Coefficients are all negative and nonsignificant. Moreover, I can reject in all cases a positive impact on test scores above 0.06 standard deviations.

E. Understanding the Null Effects

1. Did the reform change choice sets for eligible parents?

Next, I consider whether the reform truly expanded choice sets for eligible parents. Section II.B.1 shows that getting access to the extra voucher amount should have allowed students to attend schools that were, on average, higher performing. However, previous studies have shown that distance can be a major barrier for parents when choosing a school, particularly for young children. If these schools are not within reach of eligible students, then it is possible that the policy did not change students' choice sets significantly.

To get a sense of how the policy expanded choice sets for eligible students, I perform a simple simulation exercise in which I assume that each student will choose the highest-performing school that is free and within reach. I vary my definition of "within reach," assuming that parents are willing to travel 0.3, 0.7, or 1.7 miles at most, which are in the 25th, 50th, and 75th percentiles in terms of distance traveled by parents. This comes

24. Only 0.06 percent of students in the control group have dropped out of the system, and this difference is nonsignificantly different for eligible and noneligible students.

25. Previous studies have shown that a number of schools closed during this period (Grau, Hojman, and Mizala 2018). However, only 2 percent of students in the sample enrolled in a school that was later closed. This number is the same for both eligible and noneligible students.

down to assuming that ineligible students will choose the highest-performing school among public schools and private voucher schools that charge no add-ons and that eligible students will choose the highest-performing school among public schools, private voucher schools that charge no add-ons, and private voucher schools that charge add-ons and joined the program. Using these simulated choices, I then run my model to get a sense of what we could expect to see at the discontinuity if this was indeed what was driving students' choices.²⁶

Results from this exercise can be found in Table 4. Estimates are restricted to students for whom I have exact addresses. Panels A, B, and C show results from the assumptions that parents are willing to travel 0.3, 0.7, or 1.75 miles at most. As can be seen, even if we assume that parents are willing to travel as little as 0.3 miles, we would expect the treatment to allow parents to attend schools with 0.31 standard deviations higher performance indicators. If we assume that parents are willing to travel longer distances, then the treatment would allow parents to attend schools with up to 0.4 standard deviations higher performance indicators. Choosing these higher-performing schools would mean that students would end up attending, on average, higher SES schools, schools with higher add-ons, and schools with larger class sizes.

Although it is a rough exercise, the simulation allows me to get a sense of what we could expect to observe in practice. To further check whether effects are driven by students not having access to better schools due to distance constraints, I perform a second exercise in which I estimate the effect of the program on students living in places where there was more room for improving choices. To do so, I divide my sample among: (i) students living in areas where there was at least one private voucher school that charged add-ons and joined the program that outperformed all the other public and private voucher schools that charge no add-ons to parents within a 0.7 miles radius and (ii) students for whom this was not the case.²⁷ Results can be found in Table 5. There are no major differences depending on the characteristics of the neighborhoods where students were located. If anything, it appears that effects are stronger among students living in places where there was no room for improvement within 0.7 miles. However, differences are not significant.

2. *Are results different for higher- or lower-SES parents?*

In a second exercise, I analyze whether results look different for higher- or lower-SES students. We could expect results to differ by SES, as previous studies have shown that parents' preference for school characteristics tend to vary with SES (Hastings and Weinstein 2008; Bayer, Ferreira, and McMillan 2007). Also, in the Chilean context, in which schools can implement admission policies subject to few restrictions, higher quality schools could choose to serve, among eligible students, those with higher SES, leading to a differential impact of the program.²⁸

26. In modeling parents' choices, I do not consider any school capacity constraints. Also, when the student does not have any school that is free to them within reach, I assume they will choose the closest school that is free.

27. Robustness checks to supplement the heterogeneity analysis can be found in [Online Appendix 6](#).

28. Robustness checks to supplement the heterogeneity analysis can be found in [Online Appendix 6](#).

Table 4
Potential Effects on School Choice

	School Private (1)	School Test Scores (2)	School SES (3)	School Add-On (4)	School Class Size (5)	School Distance (Miles) (6)
Panel A: Individuals Travel Less than 0.3 Miles						
$R \leq \text{cutoff}$	0.308*** (0.025)	0.305*** (0.036)	0.232*** (0.030)	13.418*** (0.960)	2.453*** (0.498)	-0.052*** (0.020)
Mean control	0.227	0.153	0.414	-0.481	29.692	0.330
Observations	12,563	12,117	12,102	12,533	12,198	12,563
Panel B: Individuals Travel Less than 0.7 Miles						
$R \leq \text{cutoff}$	0.383*** (0.024)	0.407*** (0.031)	0.349*** (0.029)	21.040*** (1.221)	3.343*** (0.497)	-0.011 (0.013)
Mean control	0.307	0.616	0.566	-0.733	30.876	0.487
Observations	12,563	12,450	12,445	12,545	12,432	12,563
Panel C: Individuals Travel Less than 1.7 Miles						
$R \leq \text{cutoff}$	0.415*** (0.023)	0.357*** (0.024)	0.301*** (0.028)	25.892*** (1.466)	4.395*** (0.505)	-0.013 (0.023)
Mean control	0.307	1.158	0.820	-0.077	31.472	1.037
Observations	12,563	12,550	12,550	12,563	12,553	12,563

Notes: Table 4 shows results from simulated choices in which I assume that each student will choose the highest-performing school that is free to them within 0.3 (Panel A), 0.7 (Panel B), and 1.7 miles (Panel C). This comes down to assuming that ineligible students will choose the highest-performing school among public schools and private voucher schools that charge no add-ons, and that eligible students will choose the highest-performing school among public schools, private voucher schools that charge no add-ons, and private voucher schools that charge add-ons and joined the program. Instrumental variable estimates from *rdrobust* (Calonico, Cattaneo, and Titiunik 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6 presents heterogeneous effects by mothers' education. In general, results show no impact on the choices made by mothers with less than a high school education or mothers with a high school education or tertiary education. There is no impact on the test scores of the chosen school, the SES of the chosen school, the average class size of the chosen school, or the distance traveled to school for any of these groups. Estimates suggest that the impact on the prices of the schools chosen by parents might be higher for students with more-educated mothers. However, estimates are not precise enough to reject the hypothesis that the impact is the same for both groups of students. Results in Columns 7–9 also show no impact on educational results for students with more- or less-educated mothers.

Table 5
Heterogeneous Effects by Neighborhood

	School Private (1)	School Test Scores (2)	School SES (3)	School Add-On (4)	School Class Size (5)	School Distance (Miles) (6)	Student Language Second Grade (7)	Student Language Fourth Grade (8)	Student Math Fourth Grade (9)
Panel A: Policy Altered Choice Sets Significantly in a 0.7 Mile Radius									
Eligible	-0.014 (0.034)	-0.031 (0.052)	-0.028 (0.039)	1.482 (2.321)	0.051 (0.608)	0.000 (0.125)	-0.011 (0.080)	-0.045 (0.077)	-0.042 (0.071)
Mean control	0.636	0.458	0.898	22.954	33.755	1.203	-0.182	0.155	0.202
Observations	6,342	6,183	6,171	6,324	6,270	6,070	5,007	4,686	4,715
Panel B: Policy Did Not Alter Choice Sets Significantly in a 0.7 Mile Radius									
Eligible	0.048 (0.036)	0.060 (0.052)	0.044 (0.043)	4.923* (2.647)	0.414 (0.686)	0.140 (0.130)	-0.006 (0.081)	0.048 (0.077)	-0.018 (0.071)
Mean control	0.520	0.445	0.805	15.772	31.900	1.445	-0.162	0.080	0.155
Observations	6,108	5,910	5,906	6,076	6,018	5,721	4,785	4,549	4,553

Notes: Table 5 shows effects on school choices and educational outcomes for: students living in areas where there is at least one private voucher school that charges add-ons and joined the program that outperforms all the other public and private voucher schools that charge no add-ons to parents, within a 0.7 mile radius (Panel A), and students for whom this is not the case (Panel B). Instrumental variable estimates from rdrobust (Calonico, Cattaneo, and Titiunik 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6
Heterogeneous Effects by Mothers' Education

School Private (1)	School Test Scores (2)	School SES (3)	School Add-On (4)	School Class Size (5)	School Distance (Miles) (6)	Student Language Second Grade (7)	Student Language Fourth Grade (8)	Student Math Fourth Grade (9)
Panel A: Mother Has Less than High School Education								
Eligible	-0.044 (0.032)	0.061 (0.048)	-0.018 (0.039)	0.162 (1.144)	-0.259 (0.743)	0.139 (0.197)	-0.041 (0.071)	-0.014 (0.063)
Mean control	0.394	0.328	0.484	6.397	27.566	1.322	-0.308	-0.035
Observations	9,999	9,731	9,711	9,947	9,855	3,250	8,958	7,941
Panel B: Mother Has High School Education or Tertiary Education								
Eligible	0.040* (0.024)	-0.024 (0.035)	0.050 (0.031)	3.734** (1.896)	-0.245 (0.484)	-0.010 (0.128)	-0.062 (0.050)	-0.032 (0.046)
Mean control	0.617	0.674	0.938	23.726	32.123	1.498	0.025	0.299
Observations	12,942	12,536	12,521	12,881	12,749	5,192	12,098	11,227

Notes: Table 6 shows effects on school choices and educational outcomes for students whose mothers have less than high school education (Panel A) and students whose mothers have a high school education or tertiary education (Panel B). Instrumental variable estimates from rdrobust (Calonico, Cattaneo, and Titiunik 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3. *Are results specific to 2012?*

In a third exercise I analyze whether results are specific to 2012. Throughout the analysis, I focus specifically on 2012 because my data on the socioeconomic ranking is much more precise for that year. However, I have data on the socioeconomic ranking for 2008, which I can use as an instrument for being eligible in 2008 and 2009. This allows me to check whether results look any different in the year the policy was implemented, as well as one year after the policy was first implemented.

Results for first-grade students in 2008 and 2009 can be found in Tables 7 and 8, respectively. In each case, Panels A, B, and C show first-stage, reduced-form, and instrumental variable results. Although noisier, there is no evidence of a significant effect on parents' choices for 2008 or 2009, which indicates that results are not specific to 2012.

4. *Are results a consequence of ineligible parents becoming eligible in the future?*

As was outlined in Section IV, the socioeconomic ranking can change from year to year. Therefore, students who were eligible (ineligible) for the voucher in 2012 could in theory become ineligible (eligible) in subsequent years. In practice, we observe that 95 percent of students who were right below the cutoff in 2012 continued to be eligible by fourth grade, meaning that almost none of them had lost their benefit. However, a significant proportion of students who were right above the threshold in 2012 became eligible in subsequent years.

It seems unlikely that parents are able to foresee that they might become eligible in the future. As described in Section II.B, the Ministry of Education determines eligibility—parents do not have to apply. Also, information on the cutoff is not made available to parents. Moreover, it is worth emphasizing that although most students who were right above the threshold became eligible in subsequent years, this number decreases exponentially to the right of the cutoff. Therefore, to the extent that parents have some doubt with respect to their distance to the cutoff, they should face much uncertainty regarding their future eligibility status. Still, if we believe that ineligible parents who are close to the threshold are able to foresee that they are likely to become eligible in the future, then this could lead them to make better choices.

To analyze whether results are driven by parents right above the threshold making better choices, I perform an additional exercise in which I alter the control group to analyze if results look any different when comparing my sample of treated students to a sample of control students who are unlikely to become eligible in the future. To do so, I follow Angrist and Rokkanen (2015) and extend my results further away from the discontinuity. The approach exploits the availability of dependent variable predictors other than the running variable to estimate the causal impact of progressive vouchers for students who are further away from the cutoff. The idea is that the link between the running variable and outcomes can be broken by conditioning on a relevant set of controls. The running variable r_i can be thought of as a function of two parts $g(x_i, \varepsilon_i)$, where x_i is observed and ε_i is not. Conditional on x_i , the only source of variation in r_i , and consequently in eligibility, is ε_i . Thus, the conditional independence assumption requires that, conditional on the observed x_i , potential outcomes are mean-independent of unobserved determinants of the running variable. The strategy used is similar to a conventional

Table 7
School Choice and Educational Outcomes in 2008

	School Private (1)	School Test Scores (2)	School SES (3)	School Add-On (4)	School Class Size (5)	Student Language Fourth Grade (6)	Student Math Fourth Grade (7)
Panel A: First Stage							
Eligible	0.282*** (0.015)	0.282*** (0.015)	0.282*** (0.015)	0.282*** (0.015)	0.282*** (0.015)	0.282*** (0.015)	0.282*** (0.015)
Mean control	0.310	0.310	0.310	0.310	0.310	0.310	0.310
Observations	18,352	18,352	18,352	18,352	18,352	18,352	18,352
Panel B: Impact of Being below the Cutoff (Reduced Form)							
$R \leq \text{cutoff}$	0.012 (0.015)	-0.046* (0.023)	-0.009 (0.018)	-0.267 (0.357)	0.154 (0.338)	0.003 (0.034)	0.005 (0.033)
Mean control	0.384	-0.048	0.728	4.175	29.927	0.102	0.061
Observations	18,324	17,837	17,747	18,322	17,974	12,995	12,987
Panel C: Impact of Being Eligible for Targeted Vouchers (IV)							
$R \leq \text{cutoff}$	0.042 (0.054)	-0.164* (0.084)	-0.034 (0.064)	-0.953 (1.271)	0.545 (1.197)	0.011 (0.119)	0.019 (0.114)
Mean control	0.375	0.004	0.737	4.409	29.654	0.100	0.053
Observations	18,324	17,837	17,747	18,322	17,974	12,995	12,987

Notes: Table 7 shows effects on school choices and educational outcomes for first grade students in 2008. Panel A contains first-stage estimates, Panel B contains reduced-form estimates, and Panel C contains instrumental variable estimates, where the discontinuity is used as an instrument for being eligible for progressive vouchers. Results from robust (Calónico, Cattaneo, and Titiunik 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8
School Choice and Educational Outcomes in 2009

	School Private (1)	School Test Scores (2)	School SES (3)	School Add-On (4)	School Class Size (5)	Student Language Fourth Grade (6)	Student Math Fourth Grade (7)
Panel A: First Stage							
Eligible	0.166*** (0.013)	0.166*** (0.013)	0.166*** (0.013)	0.166*** (0.013)	0.166*** (0.013)	0.166*** (0.013)	0.166*** (0.013)
Mean control	0.670	0.670	0.670	0.670	0.670	0.670	0.670
Observations	19,004	19,004	19,004	19,004	19,004	19,004	19,004
Panel B: Impact of Being below the Cutoff (Reduced Form)							
$R \leq \text{cutoff}$	0.017 (0.015)	-0.021 (0.022)	0.023 (0.018)	-0.015 (0.354)	0.263 (0.331)	-0.013 (0.034)	0.016 (0.032)
Mean control	0.403	0.052	0.705	4.579	29,038	0.073	0.068
Observations	18,969	18,369	18,346	18,969	18,597	13,526	13,519
Panel C: Impact of Being Eligible for Targeted Vouchers (IV)							
$R \leq \text{cutoff}$	0.100 (0.088)	-0.124 (0.131)	0.134 (0.105)	-0.091 (2.111)	1.574 (1.984)	-0.067 (0.182)	0.085 (0.172)
Mean control	0.373	0.161	0.592	4.790	28,403	0.126	0.018
Observations	18,969	18,369	18,346	18,969	18,597	13,526	13,519

Notes: Table 8 shows effects on school choices and educational outcomes for first grade students in 2009. Panel A contains first-stage estimates, Panel B contains reduced-form estimates, and Panel C contains instrumental variable estimates, where the discontinuity is used as an instrument for being eligible for progressive vouchers. Results from rdrobust (Calonico, Cattaneo, and Titiunik 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

matching strategy, in which the conditional independence assumption helps to break the link between treatment status and potential outcomes. However, the approach represents an improvement over the conventional matching strategy because it uses the information inherent in the RDD to guide the choice of the conditional vector x_i and to test the veracity of the conditional independence assumption.

The approach developed by Angrist and Rokkanen (2015) proves useful in my setting, in which I have both rich information on parents' socioeconomic characteristics, including information on per-capita income used to construct the socioeconomic ranking (that is not available in previous studies) and information on the running variable. My estimates show that the conditional independence assumption is not satisfied in the sample as a whole, which is why a conventional matching strategy would lead to biased estimates. However, it is possible to break the link between the running variable and potential outcomes when I focus on a subsample that is close to the discontinuity.

The strategy allows me to compare my treatment group with a broader control group that is unlikely to become eligible in the future. Although being below the threshold increases the number of years that students have been eligible by fourth grade by just 0.7 for students who are right at the cutoff, it increases the number of years that students have been eligible by fourth grade by two years when I use this broader sample of students. It also allows me to extend my results somewhat further away from the cutoff (although it is not possible to estimate the effect on students who are far away from the threshold).

In practice, my conditional vector includes controls for mothers' education, fathers' education, household income, municipality of residence, and whether the household had access to internet and computer. Using a set of tests, I am able to show that these controls help to break the link between the running variable and outcomes for students who are in a window of $-3,000$ to $3,000$ points around the cutoff (see [Online Appendix 7](#)). Taking this conditional vector, I estimate my results using linear reweighting and propensity score weighting. Results for the linear reweighting estimator are based on Kline (2011). Kline's reweighting estimator begins with linear models for conditional means, which can be written:

$$(3) \quad E[y_i | x_i, D_i = 0] = x_i' \beta_0$$

$$(4) \quad E[y_i | x_i, D_i = 1] = x_i' \beta_1$$

This leads to the following matching style estimator at specific running variable values:

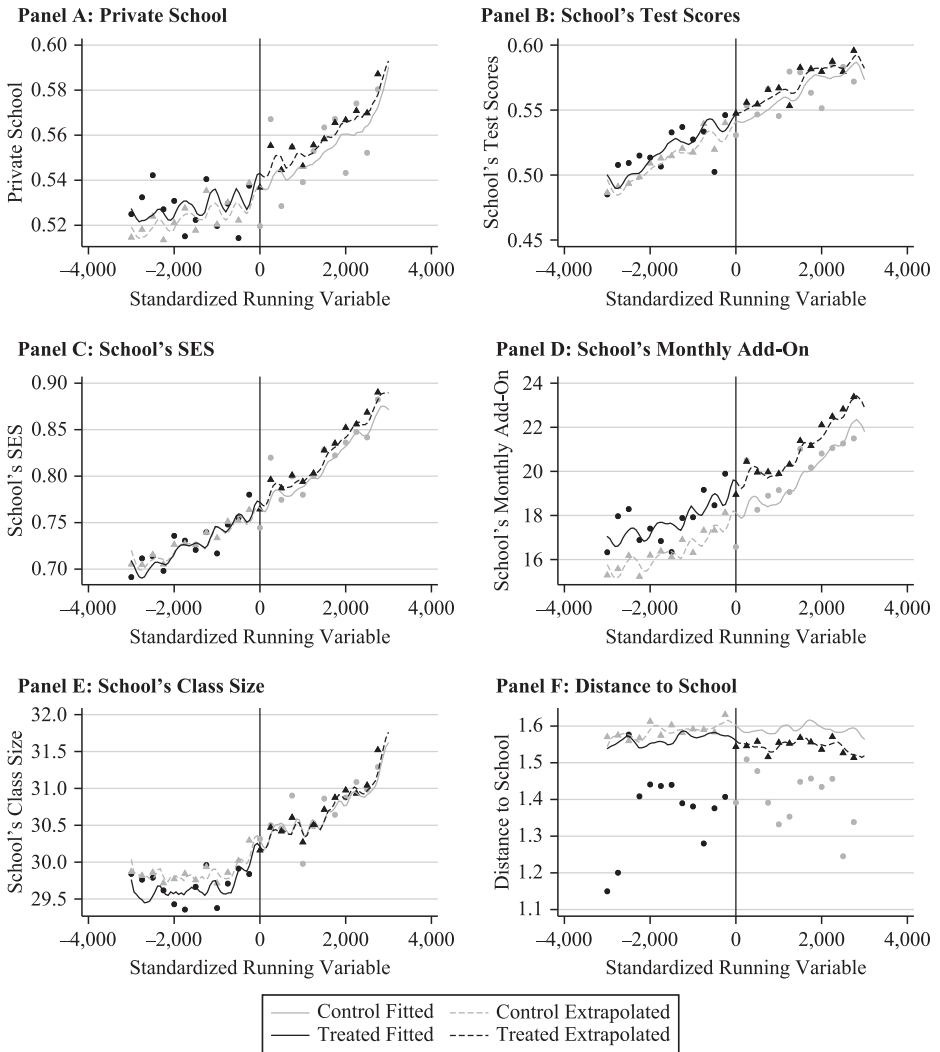
$$(5) \quad E[Y_{1i} - Y_{0i} | r_i = c] = (\beta_1 - \beta_0)' E[x_i | r_i = c]$$

Results can be found in Table 9. I estimate both the average treatment effect for students who are below the cutoff (Panel A) and the average treatment effect for students who are above the cutoff (Panel B). In each case I estimate the effect of being below the cutoff, rather than the effect of being eligible. Therefore, coefficients must be thought of as reduced-form estimates. Consistent with previous results, linear reweighting estimates and propensity score estimates are both small in magnitude, indicating that being below the cutoff for progressive vouchers did not have an impact

Table 9
Conditional Independence Results

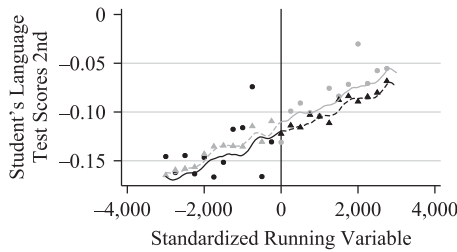
	School Private (1)	School Test Scores (2)	School SES (3)	School Add-On (4)	School Class Size (5)	School Distance (Miles) (6)	Student Language Second Grade (7)	Student Language Fourth Grade (8)	Student Math Fourth Grade (9)
Panel A: Below the Cutoff									
Linear reweighting	0.005 (0.004)	0.007 (0.006)	-0.002 (0.006)	1.365*** (0.254)	-0.207** (0.094)	-0.023 (0.042)	-0.006 (0.010)	-0.001 (0.010)	0.001 (0.009)
<i>N</i> untreated	25,732	24,995	24,954	25,604	25,329	10,239	23,490	21,387	21,420
<i>N</i> treated	24,969	24,207	24,176	24,856	24,614	10,056	23,207	21,337	21,402
Reweighting	-0.003 (0.004)	0.003 (0.006)	-0.016*** (0.006)	0.717*** (0.257)	-0.373*** (0.098)	-0.027 (0.026)	-0.010 (0.010)	-0.006 (0.010)	-0.001 (0.009)
<i>N</i> untreated	25,730	24,993	24,952	25,602	25,327	10,190	23,489	21,384	21,417
<i>N</i> treated	24,966	24,202	24,171	24,853	24,611	10,033	23,200	21,336	21,400
Panel B: Above the Cutoff									
Linear reweighting	0.006 (0.004)	0.007 (0.006)	0.009 (0.005)	1.220*** (0.310)	0.017** (0.089)	-0.047 (0.037)	-0.013 (0.009)	-0.003 (0.010)	0.007 (0.009)
<i>N</i> untreated	25,732	24,995	24,954	25,604	25,329	10,239	23,490	21,387	21,420
<i>N</i> treated	24,969	24,207	24,176	24,856	24,614	10,056	23,207	21,337	21,402
Reweighting	0.010 (0.004)	0.008 (0.006)	0.015*** (0.005)	1.553*** (0.335)	0.057*** (0.089)	-0.020 (0.024)	-0.010 (0.009)	-0.001 (0.010)	0.009 (0.009)
<i>N</i> untreated	25,730	24,993	24,952	25,602	25,327	10,190	23,489	21,384	21,417
<i>N</i> treated	24,966	24,202	24,171	24,853	24,611	10,033	23,200	21,336	21,400

Notes: Table 9 estimates average treatment effect for students who are below the cutoff (Panel A) and the average treatment effect for students who are above the cutoff (Panel B) using Angrist and Rokkanen (2015). Estimates include students who are in a window of -3,000 to 3,000 points around the cutoff.

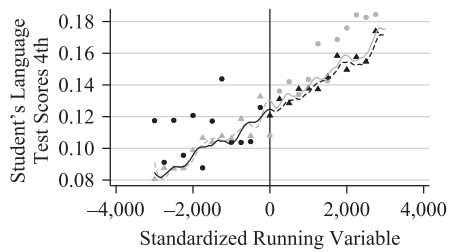
**Figure 7****Conditional Independence Assumption–Based Estimates**

Notes: This figure shows results when we extend estimates further away from the discontinuity. Gray lines show estimates of what the outcomes would be for students had they been above the cutoff ($E[Y_{0i} | r_i = c]$), and black lines show estimates of what the outcomes would be for students had they been below the cutoff ($E[Y_{1i} | r_i = c]$). Estimates of $E[Y_{0i} | r_i = c]$ to the right side of the cutoff are fitted values from regression models for observed outcomes (solid line), and estimates of $E[Y_{0i} | r_i = c]$ to the left side of the cutoff are an extrapolation based on Equation 3 (dashed line). Instead, estimates of $E[Y_{1i} | r_i = c]$ to the left side of the cutoff are fitted values from regression models for observed outcomes (solid line), and estimates of $E[Y_{1i} | r_i = c]$ to the right side of the cutoff are an extrapolation based on Equation 3 (dashed line).

Panel G: Student's Second Grade Language Score



Panel H: Student's Fourth Grade Language Score



Panel I: Student's Fourth Grade Math Score

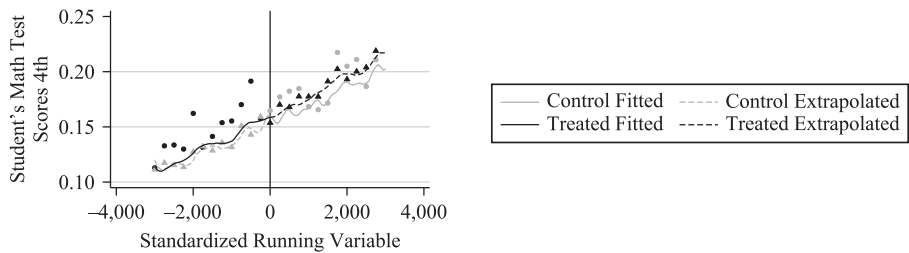


Figure 7 (continued)

on parents' choices. In line with previous findings, results show that progressive vouchers could lead parents to choose schools with higher add-ons. There are also some significant differences in schools' SES and schools' class size. However, coefficients are very small in magnitude, suggesting that access to the extra voucher could lead parents below (above) the threshold to choose schools with 0.015 standard deviations lower (higher) SES status, that it could lead parents below the threshold to reduce the class size of the chosen school by 1 percent, and that it could lead parents above the threshold to increase the class size of the chosen school by 0.2 percent.

Figure 7 provides a visual evaluation of previous results by plotting linear reweighting estimates of $E[Y_{0i} | r_i = c]$ and $E[Y_{1i} | r_i = c]$ for all values of c . In Figure 7, gray lines show estimates of what the outcomes would be for students had they been above the cutoff ($E[Y_{0i} | r_i = c]$), and black lines show estimates of what the outcomes would be for students had they been below the cutoff ($E[Y_{1i} | r_i = c]$). Estimates of $E[Y_{0i} | r_i = c]$ to the right side of the cutoff are fitted values from regression models for observed outcomes (solid line), and estimates of $E[Y_{0i} | r_i = c]$ to the left side of the cutoff are an extrapolation based on Equation 3 (dashed line). Instead, estimates of $E[Y_{1i} | r_i = c]$ to the left side of the cutoff are fitted values from regression models for observed outcomes (solid line), and estimates of $E[Y_{1i} | r_i = c]$ to the right side of the cutoff are an extrapolation based on Equation 3 (dashed line). The conditional means in the figures were constructed by plugging individual values of x_i into Equation 3 and smoothing the results using local linear regression. The figures present a picture of what the effects could look like further away from the cutoff. Consistent with estimates reported in Table 9, there are small effects along all measured outcomes for different values of the running variable.

VI. Conclusion

Much of the debate over school vouchers revolves around the idea that voucher systems may lead to high levels of socioeconomic stratification. This is undesirable from a public perspective because socioeconomic segregation typically conveys school inequities and a loss in social cohesion. School inequities arise because schools with higher-SES students benefit from positive peer effects, higher quality teachers, more involved parents, and more economic resources. At the same time, a loss in social cohesion occurs because segregation prevents students from different SES levels from sharing a common experience in schools.

The previous literature does not answer whether and to what extent alternative voucher designs can help to overcome the socioeconomic segregation that is typically associated with voucher programs. Here I address this question by looking at a reform in Chile under which voucher amounts were increased by 50 percent for students in the lowest 40 percent of the income distribution. Using a unique data set, I am able to implement a regression discontinuity design by exploiting that eligibility for progressive vouchers in Chile is a discontinuous function of a socioeconomic ranking. This allows me to estimate the impact that being eligible for progressive vouchers had on students' school choices, and their consequential distribution across schools, as well as on eligible students' educational results.

Two important conclusion can be drawn from this paper. First, eligible students did not respond to the policy by choosing schools with significantly better observable characteristics. Second, educational results did not improve in the eligibility margin, which could be a result of schools not devoting more resources to eligible as opposed to noneligible students or resources not being used in a way that improves outcomes for eligible students more than the outcomes for noneligible students.

These findings are extremely relevant from a public policy perspective. The 2008 Chilean reform is one of the most important educational reforms implemented in Chile. Moreover, the government has continued its efforts to reduce the potential barriers that keep low-income students from attending better-performing private voucher schools. Recently, the government has implemented a centralized system of school admission to prevent schools from selecting their students, and a separate plan is being undertaken by the government to completely eliminate add-ons charged by some private voucher schools to parents. Although some features of the progressive voucher policy could certainly be improved (such as enhancing private voucher school participation or providing more information to parents), it is clear from previous results that it is not just distance and price barriers that prevent parents from sending their children to better-performing schools. Therefore, it is unclear the extent to which ongoing reforms might help to reduce socioeconomic stratification significantly. Although I cannot pinpoint exactly what is preventing parents from sending their children to higher-performing schools, it seems that further action is needed to truly decrease socioeconomic segregation in an already consolidated voucher system.

Although the program did not help reduce socioeconomic stratification, existing evidence suggests that it could have led to increased educational outcomes for low-income students (MINEDUC 2012; Correa, Parro, and Reyes 2014; Villarroel 2012; Mizala and Torche 2013; Murnane et al. 2017; Neilson 2013; Navarro-Palau 2017). This work cannot determine whether the policy did or did not help to improve overall

educational outcomes. However, I provide robust evidence that, if there was indeed an improvement, it was likely driven by increased resources, increased accountability, or increased competition, and not by low-income students choosing better-performing schools. Moreover, the study shows that any improvement was likely captured by a broader population of students, regardless of whether they were eligible. This is relevant considering the efforts made by the government to target resources specifically to eligible students and have schools invest the extra resources specifically into their education. Results suggest that such efforts are fruitless and that it is probably worth relaxing such constraints because it seems unlikely that schools will target their resources towards some students and not others. Further research is needed to determine whether it was increased accountability and resources or increased competition that led to better educational outcomes. To the extent that educational improvements are driven by the former, then improvement cannot be truly attributed to progressive vouchers, but rather, to other changes that came with this program.

More broadly speaking, results are also highly relevant for a number of other countries considering an expansion of school choice systems. Empirical studies in Chile have shown how voucher systems can lead to increased segregation by SES (Hsieh and Urquiola 2003, 2006; McEwan, Urquiola, and Vegas 2008). This study takes a step further and shows that without further intervention, alternative voucher designs, such as progressive vouchers, may prove to be ineffective in terms of addressing the high levels of socioeconomic stratification that may arise as a result of voucher systems.

Appendix 1

Test Scores before and after Progressive Vouchers

This section reviews how test scores evolved during 2005–2012 for low- versus high-SES students. I use two alternative definitions to categorize students' SES. The first takes students with SES in the bottom 40 percent and top 60 percent as defined by their mother's education. Previous studies typically use mother's education to categorize students' SES (Neilson 2013; Feigenberg, Rivkin, and Yan 2019). In line with what previous authors have found, I show that differences between high- and low-SES students diminished from 0.69 standard deviations in 2005 to 0.49 standard deviations in 2012 under this definition (see Appendix Figure A.1 and Table A.1).

The second definition takes students with SES in the bottom 40 percent and top 60 percent as defined by their socioeconomic ranking. This allows me to compare eligible to noneligible students—something that was not possible in previous studies because data on eligibility are only available as of 2008. However, given that I have data on the socioeconomic ranking for the whole population of students in 2012, I can easily construct a measure of eligibility for previous cohorts by characterizing as eligible those students who were below the threshold for eligibility in 2012. Estimates using this second definition indicate that the difference in test scores between eligible and noneligible students decreased in this period from 0.37 standard deviations in 2005 to 0.3 standard deviations in 2012 (see Appendix Figure A.1 and Table A.1).

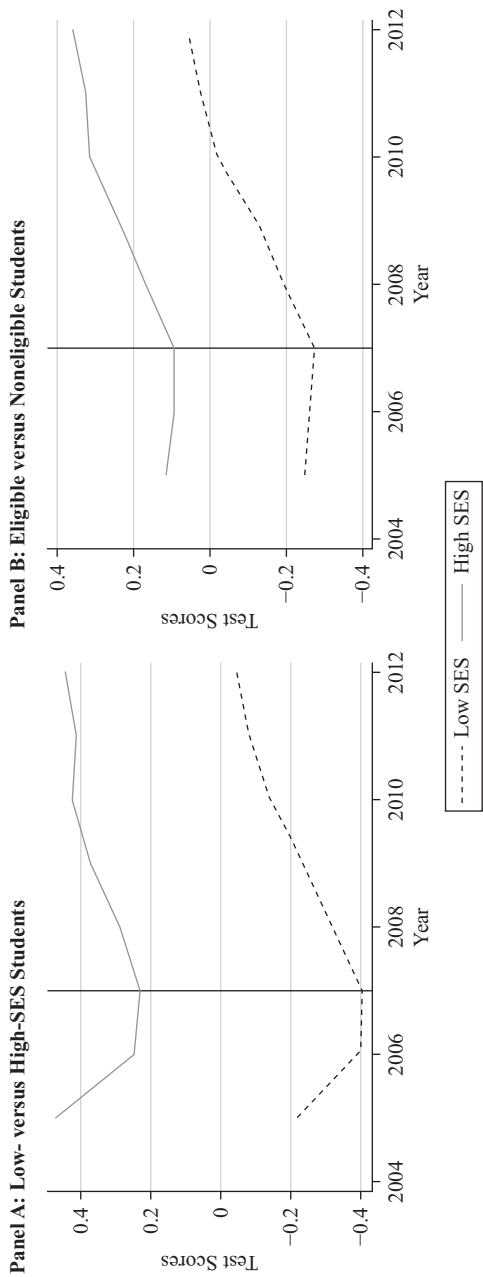


Figure A.1

Test Scores before and after Progressive Vouchers

Notes: Figure A.1 shows how test scores evolved for high and low-SES students in this period. Test scores equal the students' performance on nationwide standardized math and language tests given to students in Grade 4.

Table A.1
Test Scores before and after Progressive Vouchers

	Low SES Defined Using:	
	Mother's Education (1)	Program Eligibility (2)
Students Low-SES × 2005	−0.686 (0.010)	−0.365 (0.009)
Students Low-SES × 2006	−0.644 (0.009)	−0.357 (0.008)
Students Low-SES × 2007	−0.640 (0.010)	−0.371 (0.009)
Students Low-SES × 2008	−0.609 (0.010)	−0.364 (0.009)
Students Low-SES × 2009	−0.607 (0.010)	−0.358 (0.009)
Students Low-SES × 2010	−0.566 (0.009)	−0.334 (0.008)
Students Low-SES × 2011	−0.498 (0.009)	−0.302 (0.008)
Students Low-SES × 2012	−0.490 (0.009)	−0.301 (0.008)
Observations	1,635,198	1,835,465

Notes: Table A.1 shows how test scores evolved for high- and low-SES students in this period. Estimates are at the student level. Test scores equal the students' performance on nationwide standardized math and language tests given to Grade 4 students. Estimates include year fixed effects. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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